

In []: It **is** a sequence of characters that forms a search patterns.
It **is** used **for** search pattern to find a string **or** set of string.

Input>>text **format**>> string datatype

re>>Regular Expression

re>>RegEx>>regex

import re

In [1]: **import** re

We want to access 100 pan card data

In []: image 1 dob pan no.
image 2 dob pan no.
.
.
.
.
.
image n dob pan no.

Image

In []: Image **or** pdf **format**
PDF:-1.readable
2.Non readable(image)

1. Readable:

- 1)pdftotext
- 2)PyPdf2
- 3)PDFNumber
- 4)Tabula-py
- 5)camelet

2.Non Readable Pdf

- 1)PDF to image-pdf2image
- 2)Image to text
- 3)pytesseract(open source)
- 4)Google vision
- 5)Amazon Textract
- 6)Easy-OCR

Regular expression Functions

```
In [ ]: 1)findall()
        2)sub()
        3)match()
        4)search()
        5)compile()
        6)split()
```

1)findall()

```
In [ ]: Syntax: re.findall(pattern,text)
        This fucntion help us to find all paterrrn occurences in a string.
        It will returns list of all matches.

        pattern>>str
        text>>str
```

```
In [3]: import re
        text="""My mobile numbers are:
        9889675656 and 8899776655 and
        9970981730 and 895623784512 Todays date is 20-08-2026"""
        mob_num=re.findall(r"\b[0-9]{10}\b",text)
        mob_num
```

```
Out[3]: ['9889675656', '8899776655', '9970981730']
```

Access Date from data

```
In [4]: import re
        text="""My moble numbers are 9889675656 and 8899776655 and
        9970981730 and 895623784512 Todays date is 20-08-2026
        24-10-2026
        12/10/2026
        28/11/2026"""
        date=re.findall("[0-9]{2}[-/][0-9]{2}[-/][0-9]{4}",text)
        date
```

```
Out[4]: ['20-08-2026', '24-10-2026', '12/10/2026', '28/11/2026']
```

Access Pan card data

```
In [5]: import re
text="""Pan Card Numbers :
ASDFG1234W
POIUY3456P
JHGFD3456U
KLJHG5678I
DFGH678JH8"""

pattern="[A-Z]{5}[0-9]{4}[A-Z]{1}"
pan_card=re.findall(pattern,text)
pan_card
```

```
Out[5]: ['ASDFG1234W', 'POIUY3456P', 'JHGFD3456U', 'KLJHG5678I']
```

Access Adhar Card number

```
In [6]: import re
text="""Aadhar card numbers:
2354 4569 7895
6895 4562 1236
4561 7854 1256
"""

a_num=re.findall("[0-9]{4}\s[0-9]{4}\s[0-9]{4}",text)
a_num
```

```
Out[6]: ['2354 4569 7895', '6895 4562 1236', '4561 7854 1256']
```

Access IFSC codes

```
In [7]: import re
text="""Bank IFSC codes:
HDFC0000182
SBIN0000345
ICIC9078675"""

pattern="[A-Z]{4}[0-9]{7}"
ifsc_code=re.findall(pattern,text)
ifsc_code
```

```
Out[7]: ['HDFC0000182', 'SBIN0000345', 'ICIC9078675']
```

```
In [8]: import re
text="""Email Id
bhagyashri@gmail.com
chaitnya@yahoo.com
yashrakte12@rediffmail.com
"""

pattern="[a-z][0-9][@]{1}[a-z][.com]"
email_id=re.findall(pattern,text)
email_id
```

```
Out[8]: []
```

```
In [29]: import re
text="""Email Id
bhagyashri@gmail.com
chaitnya@yahoo.com
yashrakte12@rediffmail.com
"""

email_id=re.findall(r"[A-Za-z0-9._]+@[A-Za-z.-]+\.[a-zA-Z]{2,}",text)
email_id
```

```
Out[29]: ['bhagyashri@gmail.com', 'chaitnya@yahoo.com', 'yashrakte12@rediffmail.com']
```

```
In [28]: import re

text = """Email Id
bhagyashri@gmail.com
chaitnya@yahoo.com
yashrakte12@rediffmail.com
"""

email_id = re.findall(
    r"[A-Za-z0-9._]+@[A-Za-z.-]+\.[a-zA-Z]{2,}",
    text
)

print(email_id)
```

```
['bhagyashri@gmail.com', 'chaitnya@yahoo.com', 'yashrakte12@rediffmail.com']
```

Speacial Sequences

1.\d:String contain digit

```
In [30]: import re
text="""Aadhar card numbers:
2354 4569 7895
6895 4562 1236
4561 7854 1256
"""

a_num=re.findall("\d{4}\s\d{4}\s\d{4}",text)
a_num
```

```
Out[30]: ['2354 4569 7895', '6895 4562 1236', '4561 7854 1256']
```

2.\D

```
In [ ]: Returns a match where the string does not contain digit
```

```
In [31]: import re
text="abc1234python_23"
result=re.findall("\D+",text)
result
```

```
Out[31]: ['abc', 'python_']
```

3.\w

```
In [ ]: It returns a match the string contains any word characters
(characters for [a-z][0-9][_])
```

```
In [32]: import re
text=text = """Email Id
bhagyashri@gmail.com
chaitnya@yahoo.com
yashrakte12@rediffmail.com
"""
mail_id=re.findall("\w{7,30}@[a-z.]{5,15}",text)
mail_id
```

```
Out[32]: ['bhagyashri@gmail.com', 'chaitnya@yahoo.com', 'yashrakte12@rediffmail.com']
```

```
In [ ]:
```